

The water potential of leaf cells is affected by the water content of the soil.

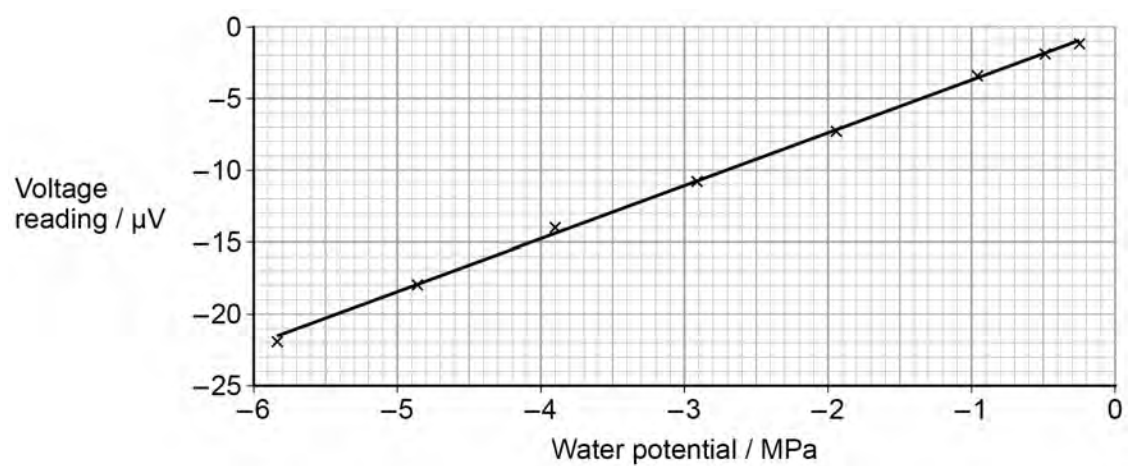
Scientists grew sunflower plants. They supplied different plants with different volumes of water.

After two days, they determined the water potential in the leaf cells by using an instrument that gave a voltage reading.

The scientists generated a calibration curve to convert the voltage readings to water potential.

Figure 8 shows their calibration curve.

Figure 8



0 8 . 1

The scientists needed solutions of known water potential to generate their calibration curve.

Table 5 shows how to make a sodium chloride solution with a water potential of -1.95 MPa

Complete **Table 5** by giving all headings, units and volumes required to make 20 cm^3 of this sodium chloride solution.

[2 marks]

Table 5

Water potential / MPa	Concentration of sodium chloride solution / mol dm^{-3}	Volume of 1 mol dm^{-3} sodium chloride solution /	
			_____ / _____
-1.95	0.04		_____

Table 6 shows some of the concentrations of sodium chloride solution the scientists used and the water potential of each solution.

Table 6

Concentration of sodium chloride solution / mol dm^{-3}	Water potential / MPa
0.04	-1.95
0.10	-4.87
0.12	-5.84

0 8 . 2

There is a linear relationship between the water potential and the concentration of sodium chloride solution.

Use the data in **Table 6** to calculate the concentration of sodium chloride solution with a water potential of -3.41 MPa

[2 marks]

Answer = _____ mol dm^{-3}

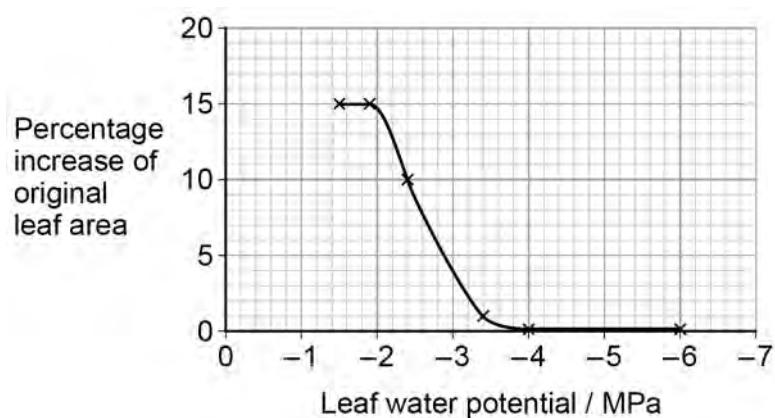
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In addition to determining the water potential in the leaf cells, the scientists measured the growth of the leaves.

They recorded leaf growth as a percentage increase of the original leaf area.

Their results are shown in **Figure 9**.

Figure 9



0 8 . 3 One leaf with an original area of 60 cm^2 gave a voltage reading of $-7 \mu\text{V}$

Use **Figure 8** (on page 28) and **Figure 9** to calculate by how much this leaf increased in area.

Give your answer in cm^2

[2 marks]

Answer = _____ cm^2

0 8 . 4

Sunflowers are not xerophytic plants. The scientists repeated the experiment with xerophytic plants.

Suggest and explain **one** way the leaf growth of xerophytic plants would be different from the leaf growth of sunflowers in **Figure 9**.

[2 marks]

0 8 . 5

Use your knowledge of gas exchange in leaves to explain why plants grown in soil with very little water grow only slowly.

[2 marks]

1

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